MATHS MANTHAN
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Subject: Mathematics **CBSE CLASS 10 – PERIODIC ASSESSMENT 11** **Time:** 60 Minutes
Class: Class X (CBSE) **Chapter 10: Circles** **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided where indicated.
2. **Section A** comprises 5 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 2 Very Short Answer (VSA) questions of 2 marks each.
4. **Section C** comprises 2 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 2 Long Answer / HOTS (LA) questions of 5 marks each.
6. **Section E** comprises 1 Applied Case-Based question of 4 marks with internal choices.

SECTION A: Multiple Choice & Assertion-Reason (6 × 1 = 6 Marks)

Q1. Quadrilateral $ABCD$ circumscribes a circle. If $AB = 9$ cm, $BC = 10$ cm, and $AD = 7$ cm, then CD equals: [1]

(A) 8 cm (B) 7 cm (C) 6 cm (D) 9 cm

Q2. If two tangents PA and PB are drawn from point P to a circle of radius 5 cm with centre O such that $OP = 13$ cm, then area of quadrilateral $PAOB$ is: [1]

(A) 30 cm^2 (B) 60 cm^2 (C) 65 cm^2 (D) 120 cm^2

Q3. If the length of a tangent from a point at distance 5 cm from the centre of a circle is 4 cm, then the diameter of the circle is: [1]

(A) 3 cm (B) 6 cm (C) 8 cm (D) 10 cm

Q4. Two concentric circles are of radii 20 cm and 12 cm. The length of the chord of the larger circle tangent to the smaller circle is: [1]

(A) 16 cm (B) 24 cm (C) 32 cm (D) 36 cm

Q5. Tangents drawn at the endpoints of a diameter of a circle are always: [1]

(A) Perpendicular (B) Intersecting (C) Parallel (D) Coincident

Q6. **Assertion (A):** If the distance between two parallel tangents to a circle is 12 cm, then the radius of the circle is 6 cm.
Reason (R): The line segment joining the points of contact of two parallel tangents is a diameter of the circle. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Type (2 × 2 = 4 Marks)

Q7. In a circle with centre O, two tangents PA and PB are drawn such that $\angle APB = 120°$. Prove that $OP = 2AP$. [2]

Q8. Two concentric circles are of radii 25 cm and 20 cm. Find the length of the chord of the larger circle which touches the smaller circle. [2]

SECTION C: Short Answer Type (2 × 3 = 6 Marks)

Q9. If $\triangle ABC$ circumscribes a circle with radius r, prove that Area$(\triangle ABC) = r \cdot s$, where s is the semi-perimeter. If area is 84 cm^2 and perimeter is 28 cm, find the inradius r. [3]

Q10. Two tangents PA and PB are drawn to a circle with centre O such that $\angle APB = 60°$. A third tangent is drawn touching the circle at C and intersecting PA, PB at D, E. Find $\angle DOE$. [3]

SECTION D: Long Answer / HOTS Type (2 × 5 = 10 Marks)

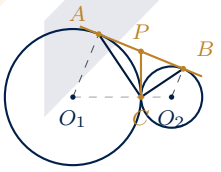
Q11. Prove that the lengths of tangents drawn from an external point to a circle are equal. Using this theorem, if $\triangle ABC$ is circumscribed about a circle of radius r touching sides BC, CA, AB at D, E, F, prove that: (i) $AF + BD + CE = AE + BF + CD = \frac{1}{2}$(Perimeter of $\triangle ABC$), (ii) Area$(\triangle ABC) = r \cdot s$. [5]

Q12. In the given figure, XP and XQ are two tangents to a circle with centre O from a point X outside the circle. ARY is a tangent to the circle at a point R intersecting XP and XQ at A and Y respectively. If $XP = 15$ cm, find the perimeter of $\triangle XAY$. If $\angle PXQ = 50°$, find $\angle AOY$. [5]

SECTION E: Case-Based / Applied Situation (1 × 4 = 4 Marks)

Q13. Gear Transmission Meshing between Tangent Pulleys

Two machine pulleys of radii $r_1 = 9$ cm and $r_2 = 4$ cm touch each other externally at point C. A direct common drive belt AB touches the first pulley at A and the second at B. A common tangent at the contact point C intersects the drive belt AB at point P.



(i) State the relation between segments PA, PB, and PC. [1]

(ii) Find the measure of angle $\angle ACB$. [1]

(iii) Calculate the total length of common tangent segment AB, OR calculate the distance between the pulley centres O_1O_2. [2]

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★★★ END OF QUESTION PAPER · ALL THE BEST ★★★